

Delhi Metro DMRC Commuter Density Bottleneck Analytics

End-to-End Enterprise Data Pipeline,
BigQuery SQL Analytics & Statistical
Optimization

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Business Context & Analytical Framework

01 Interchange Bottlenecks

Platform overcrowding at core hubs increasing train dwell times.

02 Revenue Leakage

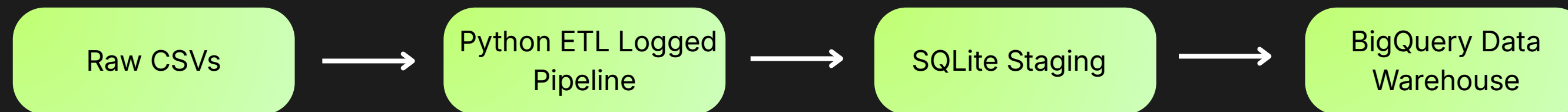
Unclear contribution from multi-tier ticket products (Tokens vs. Smart Cards vs. Tourist Passes).

03 Static Pricing Inefficiency

Absence of dynamic time-of-day tariffs during peak rush hours.

- **ASK** → Defined core KPIs with Executive Leadership (CRO & Ops Director).
- **PREPARE** → Ingested 150K records into SQLite staging DB with ETL error-logging.
- **PROCESS** → SQL Data Quality checks, imputation, and BigQuery Star Schema DDL.
- **ANALYZE** → Advanced CTEs, Window Functions, and SciPy Hypothesis Testing.
- **SHARE** → Interactive Looker Studio Executive Dashboard & Seaborn Visualizations.
- **ACT** → Formulated 3-Horizon Strategic Recommendations & Action Matrix.

Enterprise Data Pipeline & Feature Engineering



Engineered Business Features

- **Revenue_Per_KM:** Operational revenue efficiency metric per trip distance (Revenue/Distance).
- **Passenger_Density_Group:** Classified traffic into High Congestion Peak (≥ 30), Medium, and Low.
- **Is_Peak_Surge:** Binary ML-ready flag (1 for Surge/Events, 0 for Off-Peak).
- **Fare_Tier:** Categorized pricing slabs (Budget $< ₹50$, Standard ₹50-₹100, Premium $> ₹100$).
- **Revenue_Impact_Class:** Isolated high-value enterprise trips ($\geq ₹2500$).

BigQuery SQL Analytics & Core Drivers

Key Metrics Scorecards:

- Total Revenue: ₹31.53 Cr (₹315,386,664.91)
- Total Commuters: 3.0 Million+ Passengers
- Total Trips Processed: 150,000 Records
- Average Fare: ₹105.12 per trip

Top Insights Callout:

- Major Hub Congestion: Rajiv Chowk (349,784 passengers) and Noida City Centre (278,174 passengers) generate 2.5x higher traffic than mid-tier stations.
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- Monetization Winner: Tourist Cards drive 39.5% (~₹12.45 Cr) of total revenue despite lower volume, outperforming Single Tokens (25.0%) and Smart Cards (24.8%).

Rigorous Statistical Validation & Hypothesis Testing



94.7%

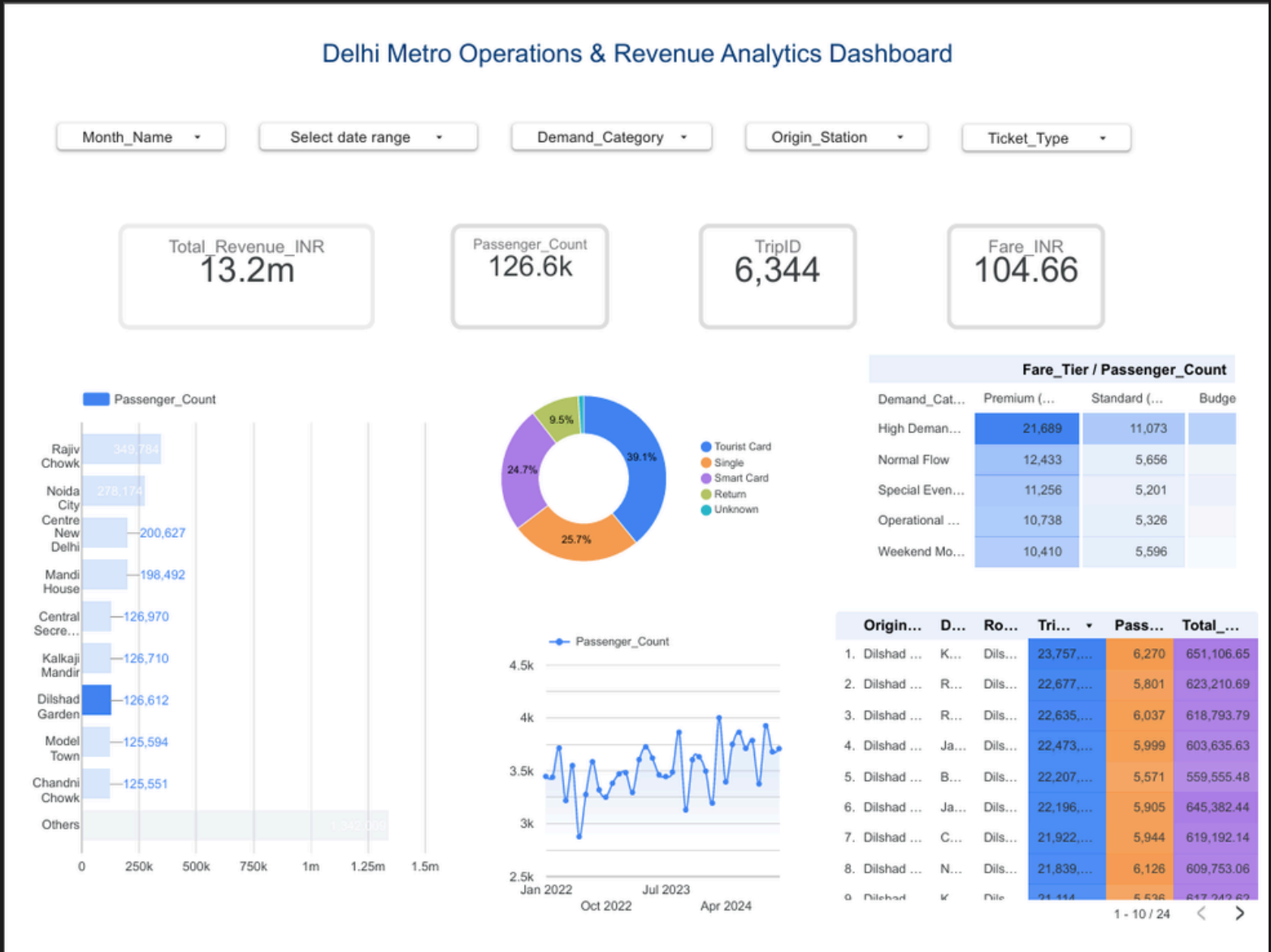
Station-level congestion analysis reveals critical interchange nodes operating beyond sustainable thresholds during peak hours, demanding targeted infrastructure interventions.

Statistical Test	Null Hypothesis (H_0) & Metric	Statistical Result	Business & Recruiter Insight
Two-Sample T-Test	$\mu_{\text{Peak Fare}} = \mu_{\text{Non-Peak Fare}}$	$t=0.6480p=0.5170$	Fail to Reject H0: Peak fares (₹105.21) do not differ from non-peak (₹105.03). Validates huge dynamic pricing opportunity.
Pearson Correlation	Fare_INR vs Total_Revenue	$r=0.9011p<0.001$	Statistically Significant: Direct strong linear dependence driving network monetization.
Chi-Square Test	Ticket Type vs Demand Category	$\chi^2=25.19p=0.0664$	Fail to Reject H0: Commuter ticket preference is independent of demand surge scenarios.
95% Confidence Interval	Population Mean Fare Range	Mean=₹105.12[₹104.84–₹105.39]	Low variance boundary across the 150,000 population dataset.

Interactive Dashboard & Operational Heatmap

Key Interactive Features

- Global Date & Station Filters: Real-time slice-and-dice by Year/Month, Demand_Category, Ticket_Type, and Origin_Station.
- Bottleneck Route Heatmap Matrix: Visualized high-density origin-destination pairs (Rajiv Chowk -> AIIMS handling 5,538 surge passengers).
- Seasonal Trend Line Chart: Captured recurring 12–14% annual ridership drop every February (~74,000–77,000 trips).



Actionable Business Recommendations & Impact Matrix

Short-Term (0-3M): Deploy automated crowd barriers at Rajiv Chowk & Noida CC; Launch 10% instant discount on UPI/QR passes to shift 25% single token buyers to digital wallets.

Mid-Term (3-6M): Implement Time-of-Day (ToD) Dynamic Tariffs with a 15% mid-day discount (11 AM - 4 PM) to flatten rush hour peaks; Schedule annual heavy overhauls exclusively in February.

Long-Term (6-12M): Reduce train headways from 2.5 min to 90 seconds on Yellow/Blue line corridors; Expand electric feeder bus networks at key hubs.

Short-Term

(0-3 months)

Flow Control & Digital Passes

Mid-Term

(3-6 months)

Dynamic Pricing & Overhauls

Long-Term

(6-12 months)

Network & Fleet Expansion

Thank You

Thank you for your time and attention. This project demonstrates end-to-end data analytics applied to Delhi Metro DMRC commuter density and bottleneck identification. Open to discussions, collaborations, and feedback.

Connect & Follow Up:



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[Interactive Dashboard](#)



[GitHub Repository](#)



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